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Chapter 4 Routing

In the simplest terms, a *router* is a device that works at layer 3 of the OSI Reference Model. Layer 3 is the network layer, which means that layer 3 is responsible for exchanging information between distinct networks. And that's exactly the function of a router: connecting two or more networks so that packets can flow freely between them.

Routing is the general term that describes what routers do. In short, when a router receives a packet from one network that is destined for a device on another network, the router determines the best way to get the packet to its destination.

In a way, routers are the post offices of the internet. When you drop a letter into a public mailbox, a mail carrier collects the mail and delivers it to a nearby post office. There, the mail is sorted and sent off to a regional post office, where the mail is sorted again and maybe sent off to yet another regional post office, and so on until the mail finally arrives at a post office close to the delivery address, where the mail is sorted one last time and given to a mail carrier who delivers the letter to the correct address.

At each step of the way, the best route to move the mail closer to its destination is determined, and the mail is sent along its way. Routers work pretty much just like that.

In this chapter, you learn about three of the most important uses for routers: connecting to the internet, connecting remote offices to each other via the internet, and managing traffic in extremely large networks such as large campuses, huge office buildings, or the internet itself.

Considering the Usefulness of Routers

Routers are an indispensable part of any network. In fact, virtually all networks require at least one router. The following sections describe the most common reasons for introducing a router into your network.

Connecting to the internet

A router is required for any network that needs access to the internet. Such a device is known as an *internet gateway* because it serves as a "gateway" to the internet.

Strictly speaking, the internet isn't a network — it's an enormous collection of millions of networks, all tied together via millions of individual routers. Your internet service provider (ISP) is just one of those millions of networks, and your internet gateway connects your private network to your ISP's network. Your ISP, in turn, provides the routers that connect the ISP's network to other networks, which ultimately connects to the internet backbone and the rest of the world.

For a home network or a very small business network, you can use an inexpensive *residential gateway* device that you can buy at a consumer electronics store such as Best Buy. A residential gateway typically includes five distinct components, all bundled into one neat package:

- A router, used to connect your private network to your ISP's network
- A small switch, typically providing from three to eight ports to connect wired devices such as computers and printers
- A wireless access point (WAP) to connect wireless devices such as laptops or smartphones
- A firewall to provide protection from intruders seeking to compromise your network
- A DHCP server to provide IP addresses for the computers and other devices on your network

Figure 4-1 shows an example of this type of setup. In this example, an ISP provides a cable feed into your house that connects to a *cable modem*, which provides a single Ethernet port to which you can connect a residential gateway. Computers on the home network are connected to the gateway's switch ports, and wireless devices connect via the Wi-Fi network. The gateway's DHCP server hands out IP addresses to any devices connected to the network.

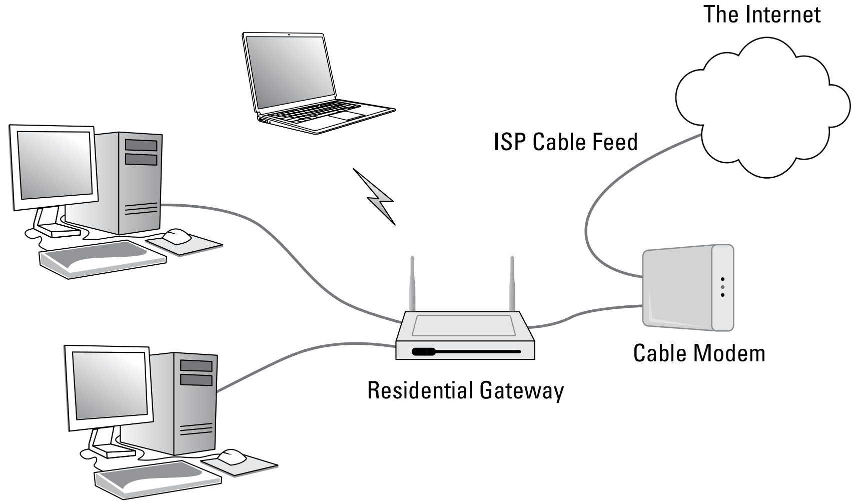

FIGURE 4-1: Connecting to the internet via a residential gateway.

Figure 4-2 shows the type of internet gateway typically used in a larger network. Here, the ISP delivers a high-speed fiber-optic feed to the customer's location and provides an *Ethernet handoff*, which is simply one or more Ethernet ports that the customer can connect to.

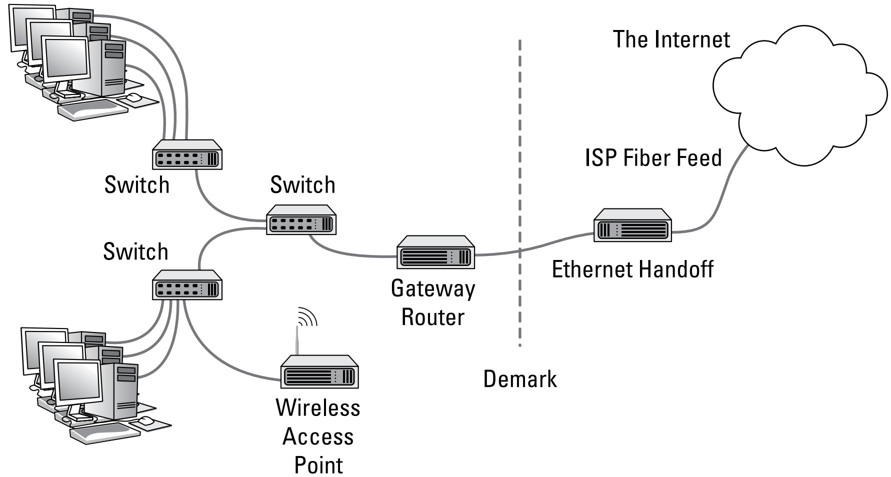

FIGURE 4-2: Connecting a larger business network to the internet.

The Ethernet handoff establishes what is called the *demarcation point*, usually called simply the *demark*. The demark is simply the dividing line that establishes who is responsible for what: The ISP is responsible for everything between the internet and the demark; the customer is responsible for everything on the private network side of the demark.

A *gateway router* is used to connect the private network to the Ethernet handoff. Much like a residential gateway, a gateway router typically provides several features into one combined device, including a router, a small switch, and a firewall. However, most business-class gateway routers don't provide Wi-Fi — the wireless network is provided by dedicated WAPs. And the small switch provided by the gateway doesn't serve the entire network; instead, it is connected to a network of switches that in turn connect the network's computers together.

One of the distinguishing characteristics of a gateway router is that it has a small number of network interfaces. At the minimum, a gateway router needs just two interfaces: an external interface and an internal interface. The *external interface,* often labeled *WAN* on the device, connects to the ISP's feed. The *internal interface* connects to the private network. (If the device includes a switch, it will have more than one internal interface.)

For more information about using a router to connect to the internet, refer to Book 3, Chapter 2.

Connecting remote locations

Another common use for routers is to connect geographically separated offices to form a single network that spans multiple locations. You can do this by using a pair of gateway routers to create a secure virtual private network (VPN) between the two networks. Each network uses its gateway router to connect to the internet, and the routers establish a secure tunnel between themselves to exchange private information.

Figure 4-3 shows how a VPN can be used to establish a site-to-site tunnel between offices in Los Angeles and Las Vegas. As you can see, each site has its own gateway router that connects to the internet. The routers are configured to provide a VPN that securely connects the two networks.

Note that the need for a VPN tunnel between networks is not related to the size of the network, at least not in terms of how many users are on the network. Instead, it's simply a function of geography. A small-town company that has two three-person offices on opposite sides of the same street can benefit from a VPN tunnel just as much as a company that has a 200-person office in Dallas and a second 200-person office in Houston.

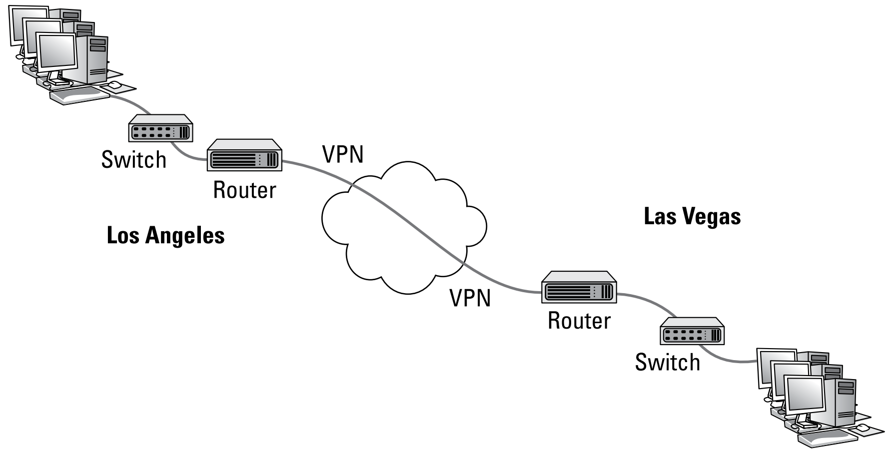

FIGURE 4-3: Connecting two networks via VPN.

Splitting up large networks

Large networks often have need for routers that are internal to the network itself. For example, consider a company that employs several thousand employees on a single campus that consists of several dozen buildings. For a network like this, routers are used to manage the network by dividing it into smaller, more manageable networks all connected with routers.

Figure 4-4 shows a simplified version of how this works. Here, a large network is segmented into two smaller networks, each on a different subnet: one on 10.0.100.*x* (subnet mask 255.255.255.0), the other on 10.0.200.*x* (also 255.255.255.0). A router is used to provide a link between the two subnets, so packets can flow from one subnet to the other.

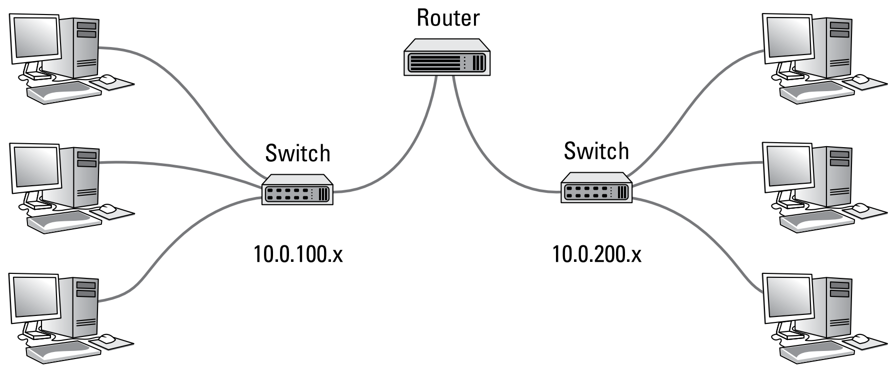

FIGURE 4-4: Using an internal router.

A router used in this way is called an *internal router* because it doesn't connect a private network to a public network. Instead, it connects two portions of a larger network.

Separate internal routers are becoming less commonplace because most switches now have routing functions built in to allow subnets to communicate with one another.

Still, very large networks still require routers to handle the large amount of traffic that must flow between networks. Picture a large college campus, in which each department has its own network each with dozens or possibly hundreds of devices. Each of those networks would have a relatively small departmental router, but a larger internal router would be used in the data center to connect all the departmental routers.

Understanding Routing Tables

Routers work by maintaining an internal list of networks that can be reached via each of the router's interfaces. This list is called a *routing table.* When a packet arrives on one of the router's interfaces, the router examines the destination IP address of the incoming packet, consults the routing table to determine which of its interfaces it should forward the packet to, and then forwards the packet to the correct interface.

Sounds simple enough.

The trick is building the routing table. In simple cases, such as a gateway router that connects a private network to the internet, the routing table is created manually with *static routes.* Configuring a gateway router with static routes isn't much more complicated than configuring a host computer with a static IP address. All you need to know is the IP address, subnet mask, and gateway address provided by your ISP for the external interface and the network address and subnet mask of the private network for the internal interface.

For more complicated environments, where multiple routers are used on the private network, special *routing protocols* are used to build *dynamic routes.* These routing protocols are designed to discover the topology of the network by finding out which routers are present on the network and which networks each router can reach.

Refer back to Figure 4-2, which depicts a small business network that has a fiber-optic connection to the internet provided by an ISP and a gateway router that connects to the Ethernet handoff.

Let's assume that the private network for this business operates on a single subnet, and the IP address for the network is 10.0.1.0 with the subnet mask 255.255.255.0. The six computers in the private network have IP addresses 10.0.101.1 through 10.0.101.6. And the internal interface on the gateway will be configured with the IP address 10.0.1.254.

Now let's assume that the ISP provides you with the following information for your Ethernet handoff:

IP address: 205.186.181.97

Subnet mask: 255.255.255.255

Default gateway: 107.0.65.31

In this example, the gateway router would have the following entries in its routing table:

Entry	Destination Network IP	Subnet Mask	Gateway	Interface
1	10.0.1.0	255.255.255.0	10.0.1.254	Internal
2	205.186.181.97	255.255.255.255	205.186.181.97	External
3	107.0.65.31	255.255.255.255	107.0.65.31	External
4	0.0.0.0	0.0.0.0	107.0.65.31	External

First, let's have a look at each of the columns in the routing table:

- **Entry:** The entry number.
- **Destination network IP:** This is the IP address of the destination network. This column is used in conjunction with the subnet mask column to determine the network to which that packet's destination IP address belongs.
- **Subnet mask:** The subnet mask that is applied to the destination IP address to determine the destination network.
- **Gateway:** The address of the router that the packet should be forwarded to.

- **Interface:** The interface that the packet should be forwarded through. Here, *internal* means the interface to which the internal private network is connected and *external* means the interface on which the ISP's handoff is connected. (In many gateway devices, these interfaces are labeled LAN and WAN, respectively.)

Now let's have a look at the four entries in this routing table:

- **Entry 1:** This entry tells the router what to do with packets whose destination is on the internal network (10.0.1.*x*). The IP address of the internal network is 10.0.1.0 and the subnet mask is 255.255.255.0. These packets will be sent to the internal interface, whose IP address is 10.0.1.254.
- **Entry 2:** This entry handles packets whose destination is the ISP's gateway (107.0.65.31). The 255.255.255.255 subnet mask means that the destination is a specific IP address, not a network. These packets are forwarded to the ISP's gateway on the external interface.
- **Entry 3:** This entry handles packets whose destination is the gateway's external interface, which has been assigned the IP address 205.186.181.97 by the ISP. These packets are forwarded to the gateway address on the external interface.
- **Entry 4:** This entry handles everything else. The network IP address 0.0.0.0 with no subnet mask means that all packets that aren't caught by any of the other rules are forwarded out to the ISP's gateway router (107.0.65.31) on the external interface.

The entries in the routing table are evaluated against each packet's destination IP address to determine where the packet should be sent. The entries are evaluated in order, and the first one that matches is used to send the packet along its way.

For example, suppose a packet is received on the external interface and the destination address is 10.0.1.5. The router will first consider entry 1, applying the subnet mask 255.255.255.0 to consider the 10.0.1.0. Because this matches the network ID in entry 1, the packet will be forwarded over the internal interface, where the switch can hand the packet off to the correct computer.

On the other hand, suppose a packet is received on the internal interface and the destination IP address is 108.211.23.42. When the router tries the first entry, the subnet mask extracts the network address 108.211.23.42. This doesn't match 10.0.1.0, so the router considers the second entry. The subnet mask 255.255.255.255

tells the router to compare the entire destination address with the IP address 107.0.65.31. Because that address doesn't match, the router tries the third entry. Again, the subnet mask 255.255.255.255 tells the router to compare the entire destination address, this time with the IP address 205.186.181.97. Again, the addresses don't match, so the router moves to the fourth and final entry in the router table. The subnet mask 0.0.0.0 reduces the entire destination address to 0.0.0.0, which matches the destination network 0.0.0.0. Therefore, the router forwards the packet on to the ISP's router at 107.0.65.31 via the external interface.

It sounds pretty simple, but in reality there is a lot more going on under the hood. In more complicated networks, there are a lot more than just four entries in the routing table. And in a busy network, a router is likely handling hundreds or even thousands of packets per second. For example, if you have 100 users on your network, all of them browsing the web, accessing email, and using other applications that cross the router, the router has an enormous workload.